TASK 1: ML CLASSIFICATION PROJECT

Breast Cancer Classification Using Machine Learning

1. Objective

The objective of this project is to build and evaluate machine-learning classification models to classify breast cancer tumours as benign or malignant. Two algorithms, Logistic Regression and Random Forest, are compared using standard evaluation metrics.

2. Tools and Technologies

- Python
- Google Colab
- Pandas
- Matplotlib and Seaborn
- Scikit-learn

3. Dataset Description

The Breast Cancer Wisconsin dataset provided by Scikit-learn was used. It contains 569 records and 30 numerical input features describing tumour characteristics. The target variable represents two classes: benign and malignant.

4. Methodology

The dataset was loaded using Scikit-learn. The data was divided into 80% training data and 20% testing data, resulting in 455 training records and 114 testing records.

Logistic Regression was implemented using feature standardization. A Random Forest classifier with 200 trees was also trained. Five-fold cross-validation was performed on the training data to assess model consistency.

5. Model Evaluation

Both models were evaluated using accuracy, precision, recall, F1-score and ROC-AUC.

Metric	Logistic Regression	Random Forest
Accuracy	98.25%	95.61%

Precision	98.61%	95.89%
Recall	98.61%	97.22%
F1-score	98.61%	96.55%
ROC-AUC	99.54%	99.31%
Five-fold CV accuracy	98.02%	96.04%

6. Results and Discussion

Logistic Regression achieved an accuracy of 98.25%, while Random Forest achieved 95.61%. Logistic Regression also achieved higher F1-score, ROC-AUC and five-fold cross-validation accuracy.

Based on these results, Logistic Regression was selected as the better-performing model for this experiment. However, both models performed well on the selected dataset.

7. Conclusion

This project demonstrated the process of loading data, splitting it into training and testing sets, training classification models, performing cross-validation and evaluating performance. Logistic Regression performed better than Random Forest on the selected evaluation measures.

This project is intended for educational purposes only. The model is not a substitute for professional medical diagnosis.

8. Project Link

GitHub repository:
https://github.com/PraneethReddy151106/InternSpark-Task1-ML-Classification

9. Evidence

Insert screenshots of the model comparison table and the confusion matrix and ROC curve plots from the Google Colab notebook.

```
Training: Logistic Regression
Training completed!

Training: Random Forest
Training completed!

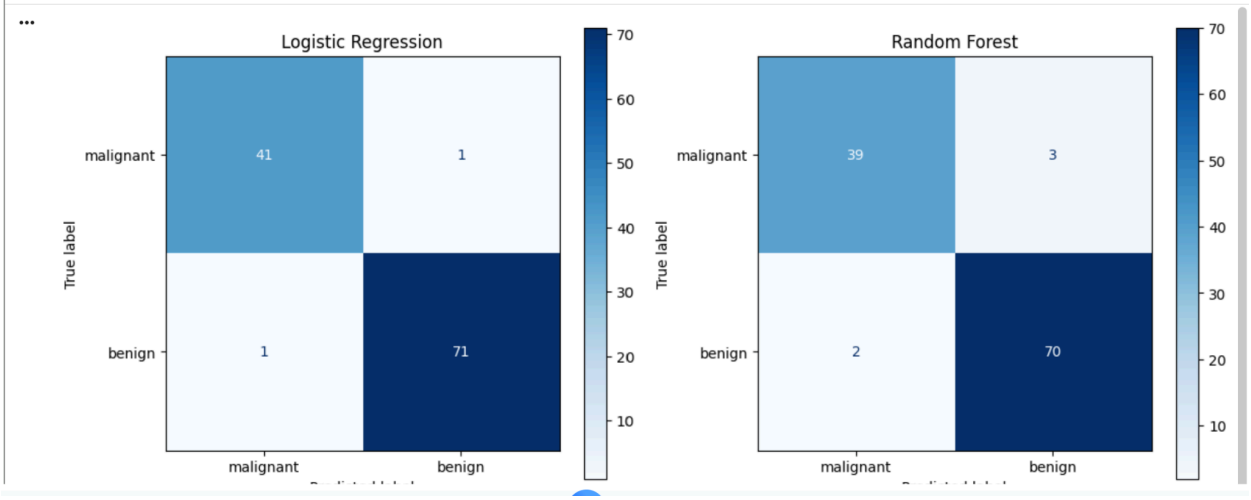
MODEL COMPARISON
```

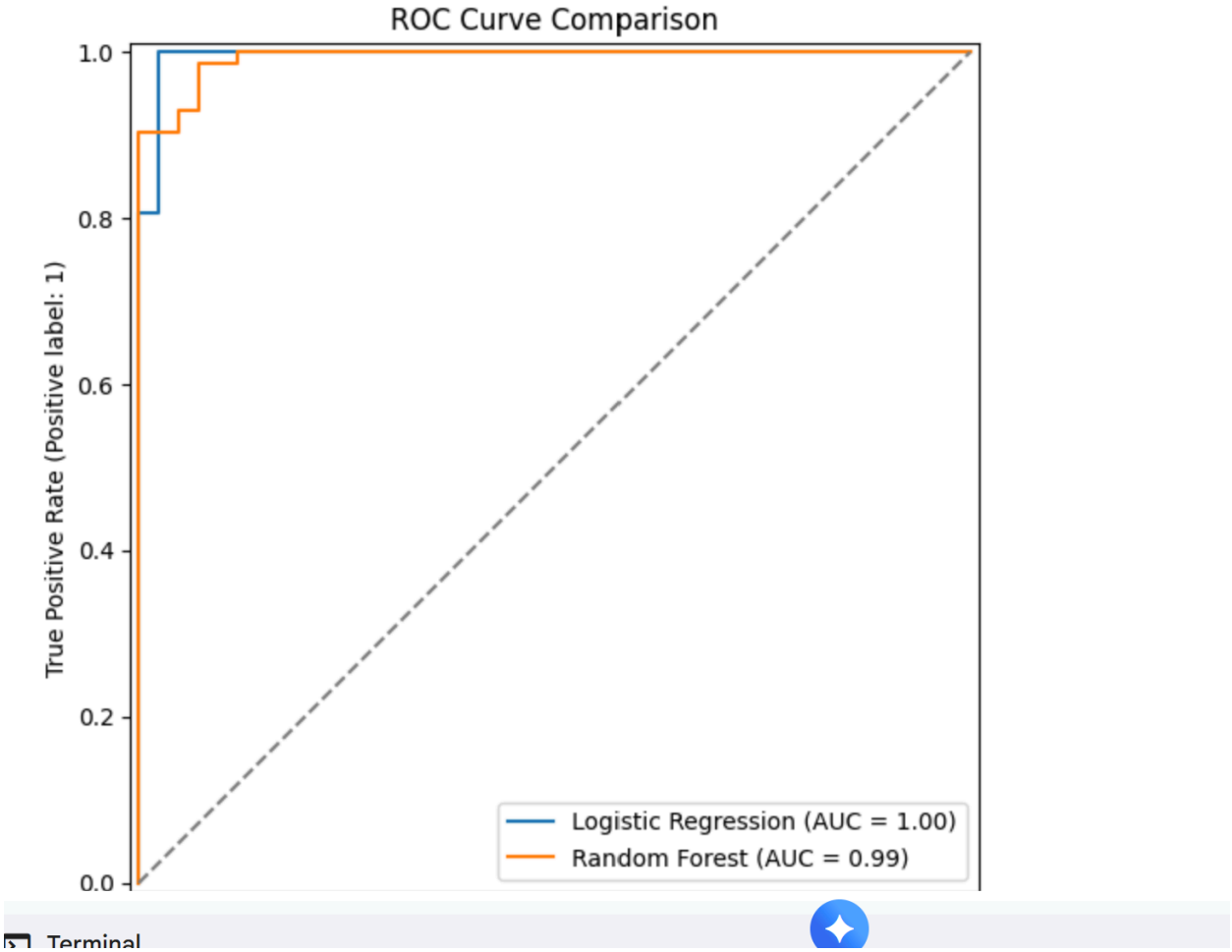
	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	5-Fold CV Accuracy
0	Logistic Regression	0.9825	0.9861	0.9861	0.9861	0.9954	0.9802
1	Random Forest	0.9561	0.9589	0.9722	0.9655	0.9931	0.9604

```
# STEP 4: Plot confusion matrices

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

for ax, (name, model) in zip(axes, models.items()):
    ConfusionMatrixDisplay.from_estimator(
        model,
        X_test,
        y_test,
        display_labels=data.target_names,
    ax.set_title("ROC Curve Comparison")
    plt.tight_layout()
    plt.show()
```




ROC Curve Comparison
True Positive Rate (Positive label: 1)
1.0
0.8
0.6
0.4
0.2
0.0
Logistic Regression (AUC = 1.00)
Random Forest (AUC = 0.99)

Terminal